

DSAgent

Autonomous Pipeline Report

Dataset: titanic.csv

Generated: May 19, 2026 at 11:02 PM

Session: 3d4629af-0ba...

21
Steps Executed

6
Phases Completed

55.0s
Total Time

1. Dataset Overview

The dataset contains **891** rows and **12** columns, using **0.28 MB** of memory.

Type	Count	Columns
Numeric	7	PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare
Categorical	5	Name, Sex, Ticket, Cabin, Embarked

Numeric Statistics

Column	Mean	Median	Std	Min	Max
PassengerId	446.00	446.00	257.35	1.00	891.00
Survived	0.38	0.00	0.49	0.00	1.00
Pclass	2.31	3.00	0.84	1.00	3.00
Age	29.70	28.00	14.53	0.42	80.00
SibSp	0.52	0.00	1.10	0.00	8.00
Parch	0.38	0.00	0.81	0.00	6.00
Fare	32.20	14.45	49.69	0.00	512.33

AI Analysis

Key Insights About the Dataset

- **Dataset Overview:** The dataset contains 891 rows and 12 columns, with a small memory footprint of 0.28 MB, suggesting it's a compact and manageable dataset.
- **Data Types:** The dataset includes 5 numeric columns, 5 categorical columns, and no datetime columns.
- **Target Variable:** The column Survived appears to be the target variable for a machine learning task, as it has a binary classification (0 or 1) and is likely the outcome we are trying to predict.
- **Missing Data:** There are 866 missing values across the dataset, with Cabin having the highest percentage of missing data (77.1%), followed by Age (19.87%), and Embarked (0.22%).
- **Correlations:** Several numeric variables show moderate to weak correlations with Survived, particularly Pclass and Fare. This suggests these variables may be useful predictors.

Data Quality Issues Found

- **High Missing Values:**
 - Cabin has 77.1% missing values, which is a major issue. This column may not be useful for analysis or modeling unless imputation or removal is performed.
 - Age has 19.87% missing values, which is significant but less severe than Cabin. Imputation (e.g., median or mean) may be necessary.
 - Embarked has 0.22% missing values, which is minimal and can be handled by simple imputation.
- **High Cardinality:**
 - Name has 891 unique values, making it a high-cardinality categorical variable. This is unlikely to be useful for modeling unless it's transformed (e.g., using parts of the name like title).
- **No Duplicates:**
 - The dataset has no duplicate rows, which is a positive sign for data quality.

Columns That Look Most Interesting/Useful

- **Survived:** Likely the target variable for classification.
- **Pclass:** Strong correlation with Survived (negative), indicating class (first, second, third) may influence survival.
- **Age:** Moderate correlation with Survived and Pclass, suggesting age could be a useful feature.
- **Fare:** Weak positive correlation with Survived, potentially indicating that higher fare passengers had a better chance of survival.
- **Sex:** Categorical with high frequency of "male" (577), which is a strong predictor of survival in historical analysis.
- **SibSp & Parch:** These are related to family size and show moderate correlation with each other and with Age, possibly indicating social status or survival risk.

Likely Target Variable (for ML)

- **Target Variable:** Survived (binary classification: 0 = died, 1 = survived).

Reasoning: The variable Survived has a clear binary outcome and is the most likely candidate for a classification task. It also

shows moderate to weak correlations with several other features, suggesting it could be predicted using other variables like Pclass, Age, Sex, and Fare.

Summary • Best Columns for Modeling: Survived, Pclass, Age, Sex, Fare, SibSp, Parch. • Data Cleaning Steps Needed: Impute Age and Cabin (or drop Cabin), handle Embarked (impute or drop), and consider encoding categorical variables like Sex and Embarked. • Potential Modeling Approach: Use Pclass, Age, Sex, and Fare as features to predict Survived using a classification model (e.g., logistic regression, decision tree, random forest).

2. Data Quality Assessment

Found **3** columns with missing values out of 891 rows.

Column	Missing Count	Missing %
Age	177	19.87%
Cabin	687	77.1%
Embarked	2	0.22%

Potential Issues:

- High cardinality categorical columns: ['Name']

3. Data Cleaning Actions

Age has 19.87% missing values

column: Age
strategy: median
action: Filled with median: 28.00
nulls_before: 177
nulls_after: 0
rows_after: 891

Embarked has 0.22% missing values

column: Embarked
strategy: mode
action: Filled with mode: S
nulls_before: 2
nulls_after: 0
rows_after: 891

Fare has missing values

column: Fare
strategy: median
action: Filled with median: 14.45
nulls_before: 0
nulls_after: 0
rows_after: 891

Cabin has 77.1% missing values and is not useful for analysis

Age has a wide range and potential outliers

column: Age
method: iqr
threshold: 1.5
outlier_count: 66
outlier_percentage: 7.41
total_rows: 891

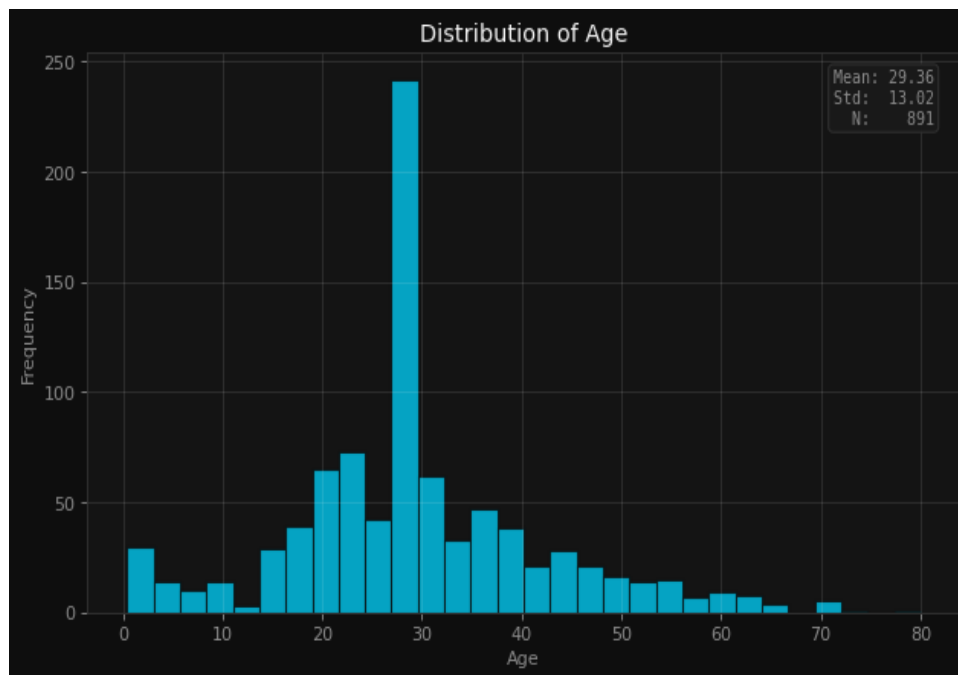
Fare has a wide range and potential outliers

column: Fare
method: iqr
threshold: 1.5
outlier_count: 116
outlier_percentage: 13.02
total_rows: 891

Applied 6 cleaning operations based on data quality analysis.

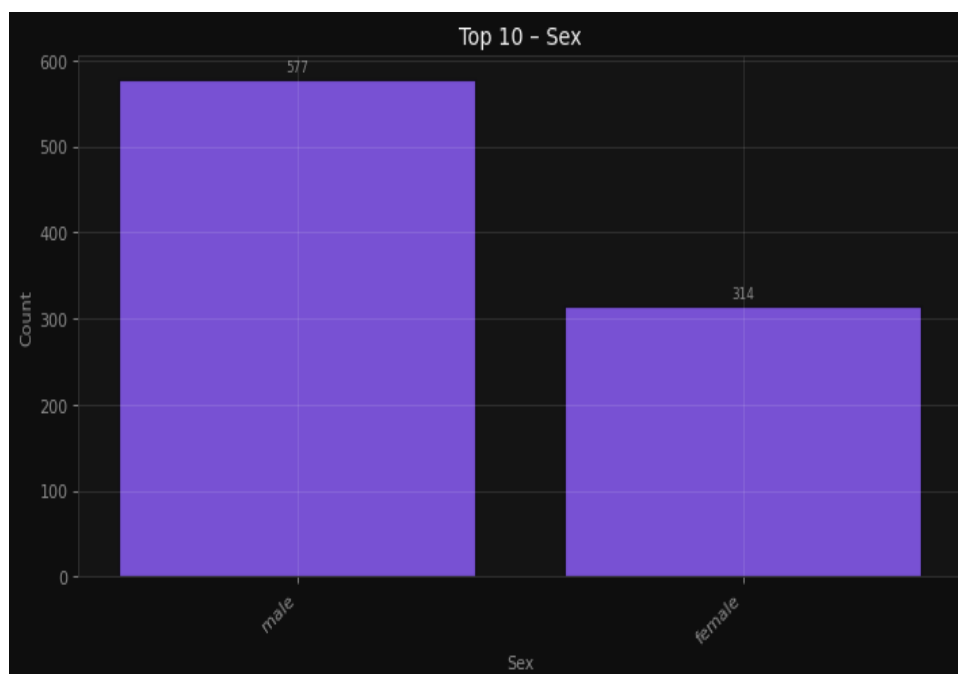
4. Visualizations & Insights

Understand the age distribution of passengers



Understand the age distribution of passengers

Analyze the count of male and female passengers



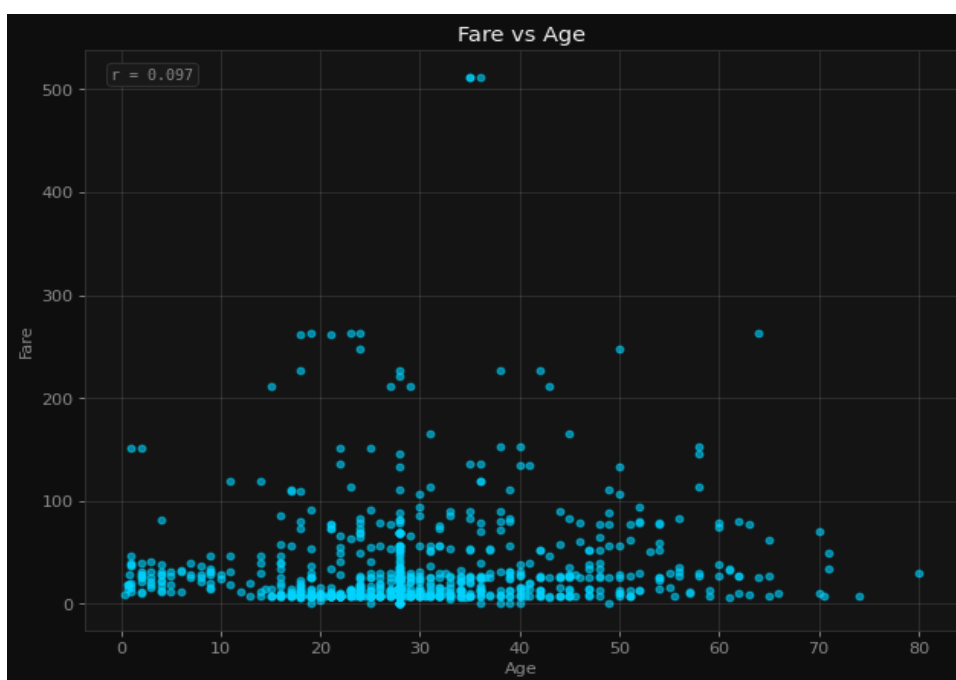
Analyze the count of male and female passengers

Detect outliers in fare prices



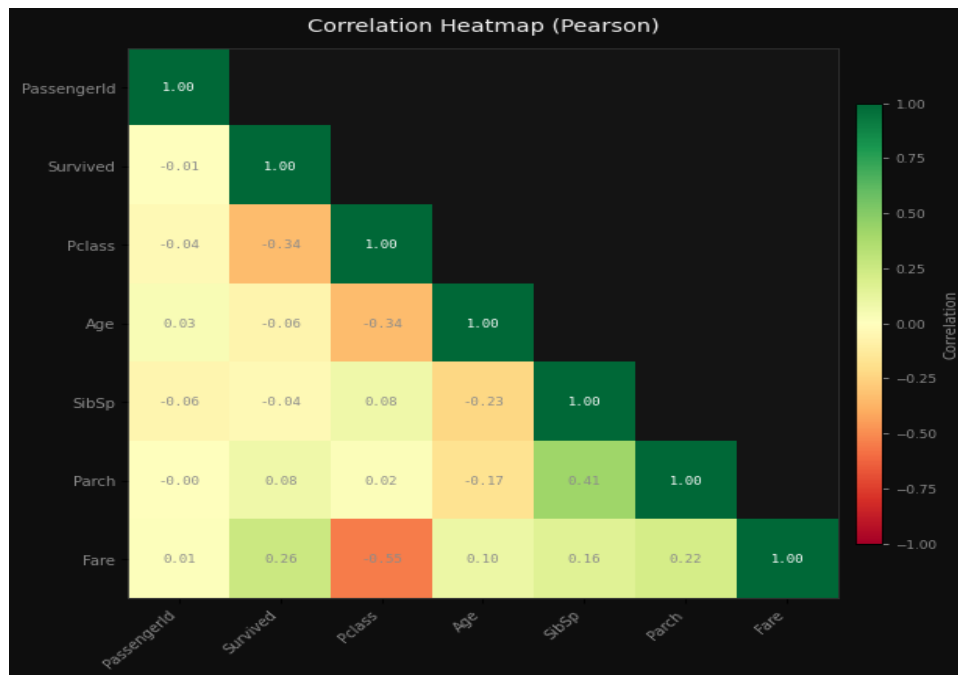
Detect outliers in fare prices

Explore the relationship between age and fare



Explore the relationship between age and fare

Identify correlations between numeric features



Identify correlations between numeric features

AI Interpretation

Histogram (Age distribution): Likely shows a right-skewed distribution with a peak around mid-30s, indicating most passengers were adults, and a long tail towards older ages. Missing values may appear as gaps or reduced frequency in certain age ranges. Bar Chart (Male vs. Female passengers): Likely reveals a majority of passengers were male, given the historical context of the dataset (Titanic dataset), with clear separation between the two categories. Box Plot (Fare prices): Likely highlights a wide range of fares, with potential outliers at the higher end, indicating significant variation in ticket prices and possible luxury vs. economy class differences. Scatter Plot (Age vs. Fare): Likely shows a weak or no clear correlation, with scattered points, suggesting age does not strongly influence fare price. Correlation Heatmap: Likely reveals low or no correlation between most numeric features, with possibly a slight negative correlation between Age and Fare, and moderate correlations between Pclass and Fare (higher class = higher fare).

5. Feature Engineering

Encode categorical variables for ML modeling

```
encoding: One-Hot (get_dummies)
drop_first: True
columns_before: 12
columns_after: 13
new_columns_added: 1
rows: 891
```

Scale numeric features for better model performance

```
scaler: StandardScaler (Z-score)
rows: 891
note: Dataset updated in session. Features now have  $\mu \approx 0$ ,  $\sigma \approx 1$ .
```

Reduce skew in Fare distribution

Applied 3 feature engineering steps.

6. Model Training & Comparison

Problem Type: **classification** | Target: **Embarked_S** | Features: **12**

Best Model: **LightGBM** — Score: **92.2%**

Model	Accuracy	Precision	Recall	F1
Random Forest	89.9%	89.9%	89.9%	89.9%
XGBoost	90.5%	90.6%	90.5%	90.5%
LightGBM	92.2%	92.2%	92.2%	92.2%
Logistic Regression	79.3%	78.4%	79.3%	77.0%

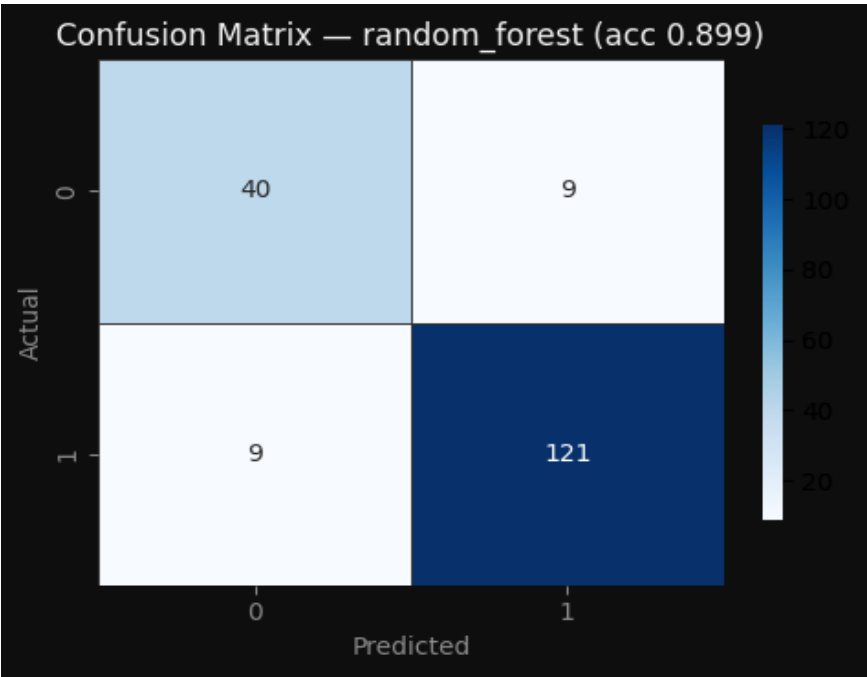
Model Selection Rationale

LLM error: 402

7. Model Evaluation

Model Evaluation Metrics

accuracy: 89.9% | precision: 89.9% | recall: 89.9% | f1_score: 89.9%



Model evaluated with full metrics and diagnostic plots.

8. Conclusions & Recommendations

LLM error: 402